Proposed Enhancements to Traffic Alert System

Rule-based & Machine Learning

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Project Deliverables Based on Requirements

To fulfill all the items listed in Milestone 2, the following actions will be taken:

Requirement	Action
Build Database to Support Traffic Accident Model	A dedicated, scalable relational database (PostgreSQL or Azure SQL) will be built and structured to support both real-time and manual data ingestion. Schema design will support historical and incoming data, incorporating accident events, location, weather, and traffic parameters.
Ensure Database is Clean, Organized and Accommodates Real-time Data from Various Sources	Data pipelines will be developed to support structured parsing, standardization, and real-time ingestion using Azure services (Azure Data Factory / Event Hub). Duplicate handling, missing values, and normalization procedures will be implemented.
Ensure that Duplicates Are Eliminated	Deduplication logic using unique keys (timestamp + coordinates + event type) and similarity measures will be applied before storing data in the master database.
Host on Azure	The complete pipeline including database, FastAPI service, and storage will be hosted and configured securely on Azure Cloud using Azure SQL Database, Azure Blob Storage, and App Service.
Ensure All Coding, Parsing and Scheduling Allows for Real-Time Data Import Efficiently to Master Database	Scripts using Python and schedulers like Azure Logic Apps or Azure Functions will be configured for real-time and periodic imports from APIs or external feeds.
Develop Necessary Tools to Allow Manual Uploading of Datasets	A secure web-based interface will be created using FastAPI + Streamlit/HTML to allow manual dataset uploads and simulate data if needed.
Integrate Data from Other Tools Such as ArcGIS to Feed Database	ArcGIS REST services will be integrated using scheduled fetchers to feed relevant spatial and traffic data directly into the main database.
Consider Expanding Model to Include Other Factors That Can Improve Overall Accuracy	As reflected below, the model already includes precipitation, temperature, visibility, and weather as predictive features.
Expand Database to Include Real-time Sources: New York, Los Angeles, Las Vegas, etc.	The pipeline will support feeds from open traffic APIs and weather datasets covering major cities including NYC, LA, and Las Vegas.

Proposed Rule-based Enhancements to Alert Prediction Pipeline

Rule-based Risk Scoring

Risk Scoring System:
● Define rules for each data feature to assign a risk score. For example:
● Weather Conditions: If the weather is "foggy", "rainy", or "snowy", increase the risk score.
● Temperature: Extremely low or high temperatures (e.g., below 0°C or above 35°C) can increase accident risk.
● Visibility: Low visibility (e.g., below 500 meters) can raise the risk of accidents.
● Speed: If the average speed of vehicles is too low (indicating congestion) or too high (indicating reckless driving), increase the risk score.
● Precipitation: Higher precipitation levels (e.g., over 10 mm/hr) should result in higher risk.

Location-based Risk:
● Continue using the grid system and apply rules to grid areas based on the data collected.
● For instance, areas with higher accident frequency and worse weather conditions can be assigned a higher risk score.
● Apply additional rules for areas near intersections, curves, or areas with poor road conditions (e.g., potholes, roadworks).

Enhanced Grid-based Alert System

Dynamic Grid Rules:
● Modify the existing grid system to factor in new data features.
● For each grid cell (representing a geographic area), calculate a combined risk score based on:
● Accident frequency (as we did previously).
● Weather severity (from weather data).
● Traffic speed (if available).
● Visibility and precipitation.

Alert Threshold:
● Set a risk score threshold that determines when an area should be flagged as a "dangerous zone".

● If the risk score for a grid exceeds the threshold (e.g., 70%), send an alert to users entering that area.
● Adjust the threshold dynamically based on the time of day or season. For example, increase the threshold for alerts at night or during winter months when accidents are more frequent.

Alert Notifications

Proximity-based Alerts:
● As the user approaches a grid with a high risk score, send an alert warning them about the potential danger. Include specific information about the weather and visibility conditions.
● Consider additional alerts for hazardous weather conditions (e.g., fog, rain) based on the user's location.

Dynamic Alerts Based on Time:
● Adjust alerts for the time of day. For instance, in the evening or at night, use a lower risk threshold because visibility tends to be reduced.
● Similarly, set different rules for weekends vs. weekdays, considering higher traffic volumes during certain days.

Potential ML Based Enhancements

Accident Risk Prediction Model:

● Train a multivariate model using machine learning algorithms (e.g., Random Forest, Gradient Boosting, or XGBoost) to predict accident risks based on the expanded features.
● Model inputs should include geographical location, time of day, weather conditions, traffic speed, and precipitation. The target variable could be accident severity or likelihood.
● Use classification models to categorize accident risk (e.g., high, medium, low) and regression models to predict the accident frequency in specific grid cells.

Real-Time Predictions:

● Implement real-time prediction using the FastAPI endpoint to process user data (location, time) and determine if the user is entering a dangerous zone based on current weather conditions, speed, and historical accident data.

Expanded Geographic Coverage for Accident Prediction

To enhance the system's effectiveness and scalability, accident risk prediction will be extended to additional key regions:

- **New York (USA)**
  Real-time traffic, weather, and accident datasets will be integrated from public APIs and regional data sources to support accurate accident risk analysis within New York City and surrounding areas.
- **Canada**
  Provincial and national-level data (e.g., from Transport Canada, Open Government Portal) will be utilized to support predictive modeling for major Canadian cities, with region-specific rules and weather patterns incorporated into the pipeline.
- **United Kingdom (UK)**
  Data from UK's Department for Transport and other open platforms will be used to map high-risk zones and perform machine learning-based accident prediction tailored to UK road infrastructure and traffic behavior.

Cost & Timeline

Item	Details
Total Cost	**$900 USD**
Timeline	**3 Weeks**
Week 1	Database setup, real-time ingestion pipeline, and Azure deployment
Week 2	Rule-based engine implementation, grid scoring logic, and ArcGIS integration
Week 3	Manual upload interface, ML model training, API integration, and rule-based setup for accident prediction in New York, UK, and Canada

Closing Note

Thank you for considering this proposal. I look forward to the opportunity to bring this project to life and am committed to delivering it with the highest standards of quality and professionalism. Please feel free to reach out for any clarifications or adjustments.